CS23334 Fundamentals of Data Science Lab

Experiment 9

Understanding Matplotlib & Seaborn Library Use Cases in Data Science

Objective: To explore, analyze, and visualize multi-dimensional feature distributions and relationships in the Iris dataset using Matplotlib and Seaborn.

Methodology:

1. **Data Ingestion & Inspection:** Load the Iris dataset and inspect summary statistics, data types, and variety distributions.
2. **Categorical & Bivariate Visualization:** Use count plots and scatter plots to analyze class balances and dimension separations.
3. **Multivariate Pairplot:** Generate pairwise scatter and kernel density plots to capture multi-attribute clustering patterns.
4. **Faceted Histograms:** Construct conditional distribution grids across all sepal and petal features grouped by variety.

Code:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.read_csv('Iris_Dataset.csv')

print(data.info())
print(data.describe())
print(data.value_counts('variety'))

sns.countplot(x='variety', data=data)
plt.show()

dummies = pd.get_dummies(data.variety)
FinalDataset = pd.concat([pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]], axis=1)
print(FinalDataset.head())

sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='variety', data=data)
plt.show()

sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm', hue='variety', data=data)
plt.show()

sns.pairplot(data, hue='variety', height=3)
plt.show()

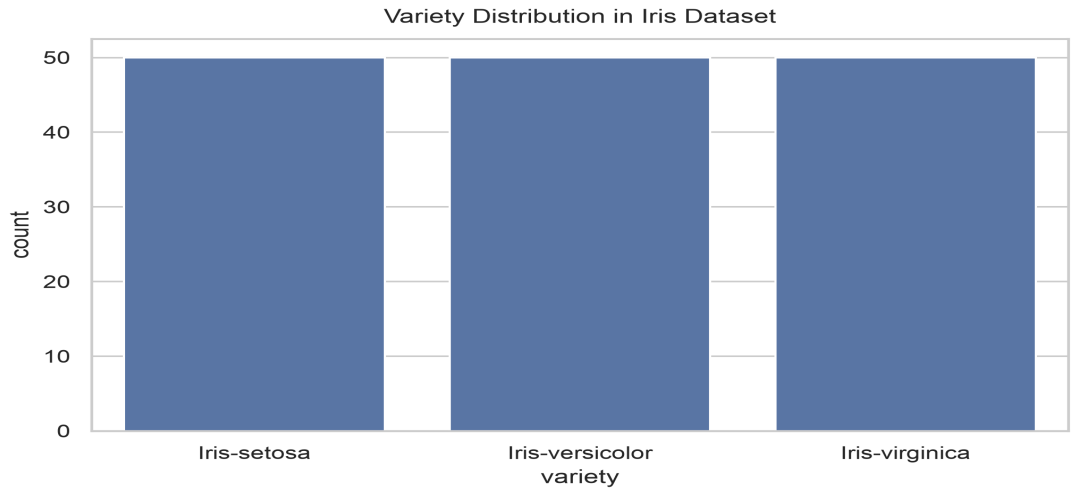
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'PetalLengthCm').add_legend()
plt.show()

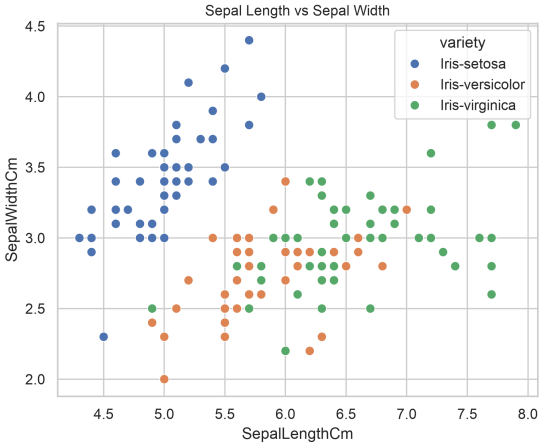
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'PetalWidthCm').add_legend()
plt.show()

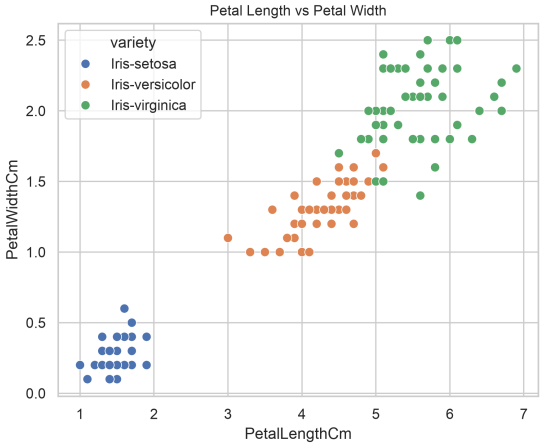
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'SepalLengthCm').add_legend()
plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'SepalWidthCm').add_legend()
plt.show()
```

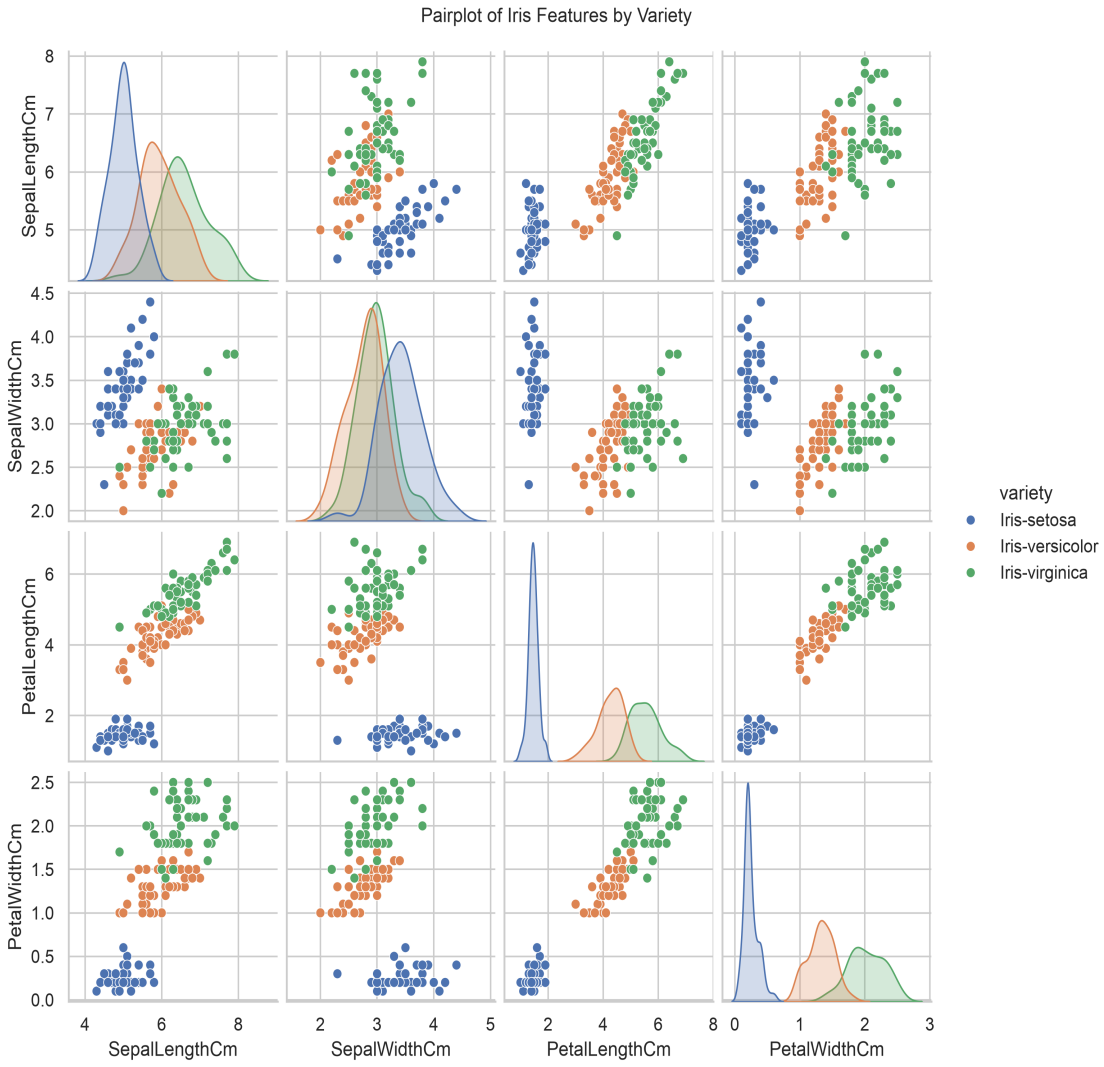
OUTPUT: 1. Categorical Distribution & Bivariate Scatter Plots



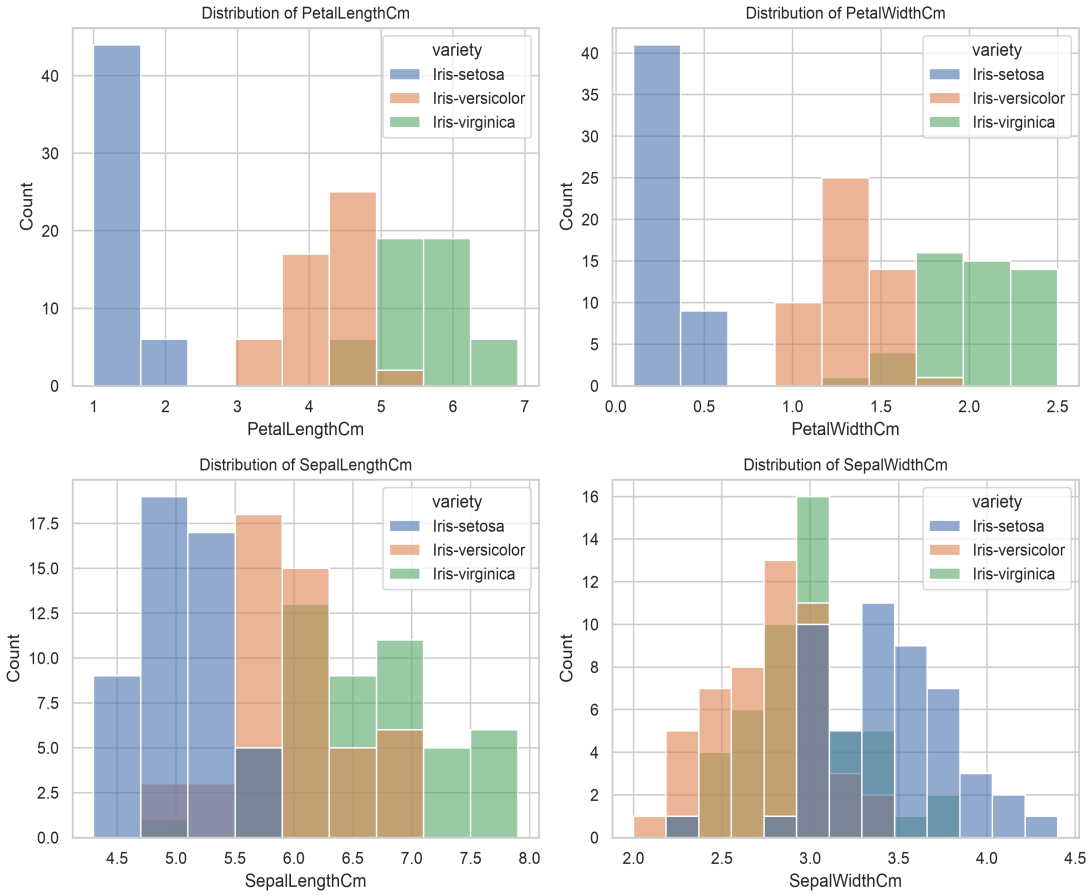




OUTPUT: 2. Multivariate Pairplot Analysis



OUTPUT: 3. Feature Distributions (FacetGrid Histograms)



Result:

Matplotlib and Seaborn visualizations on the Iris dataset successfully revealed distinct class separability. Pairplots showed that Setosa is linearly separable from Versicolor and Virginica, while heatmaps confirmed strong positive correlation between petal length and petal width.